

SQL综合查询



引言

一个完整的数据查询语句的格式

SELECT [ALL|DISTINCT] <目标列表达式1> [, <目标列表达式2> , ...]

FROM <表名或视图名1> [, <表名或视图名2> , ...]

[WHERE <元组选择条件表达式>]

[GROUP BY <属性列名1> [, <属性列名2> , ...] [HAVING <组选择条件表达式>]]

[ORDER BY <目标列名1> [ASC|DESC] [, <目标列名2> [ASC|DESC] , ...]]



数据库实例

学校教务管理系统数据库

- 学生表STUDENT (学号SNO, 姓名SNAME, 性别GENDER, 所在班级号CNO)
- 班级表CLASS (班级号CNO, 所在院系DEPARTMENT, 所属专业SPECIALITY, 班长学号MONITOR)
- 课程表LESSON (课程号LNO, 课程名LNAME, 教材名BOOK, 学分CREDIT)
- 教师表TEACHER (教师编号TID, 姓名TNAME, 所在院系DEPARTMENT)
- 班级选课表ELECTION (班级号CNO, 课程号LNO, 教师编号TID, 上课年度SYEAR, 上课学期SEMESTER)
- 学生成绩表GRADE (学生学号SNO, 课程号LNO, 分数SCORE)

注意：表中CREDIT, SCORE, YEAR属性为INT类型，其余为CHAR类型。



练习1

学生表STUDENT (学号SNO, 姓名SNAME, 性别GENDER, 所在班级号CNO)

班级表CLASS (班级号CNO, 所在院系DEPARTMENT, 所属专业SPECIALITY, 班长学号MONITOR)

课程表LESSON (课程号LNO, 课程名LNAME, 教材名BOOK, 学分CREDIT)

教师表TEACHER (教师编号TID, 姓名TNAME, 所在院系DEPARTMENT)

班级选课表ELECTION (班级号CNO, 课程号LNO, 教师编号TID, 上课年度SYEAR, 上课学期SEMESTER)

学生成绩表GRADE (学生学号SNO, 课程号LNO, 分数SCORE)

```
SELECT monitor, sname, speciality, class.cno
FROM class, student
WHERE class.monitor = student.sno;
```

100 %

结果 消息

	monitor	sname	speciality	cno
1	s01	王玲	大数据	1
2	s04	赵泽	人工智能	2
3	s05	崔雪	计算机	3
4	s07	王明	计算机	4

查询所有班长的学号，姓名，所在班级号和所学专业。

```
SELECT MONITOR,SNAME,CLASS.CNO, SPECIALITY
FROM CLASS, STUDENT
WHERE CLASS .MONITOR = STUDENT.SNO;
```



练习2

- 查询2018年度讲授过两门或两门以上课程的教师编号和所教授的课程号。

```
SELECT DISTINCT E1.TID,E1.LNO  
FROM ELECTION AS E1,ELECTION AS E2  
WHERE E1.SYEAR=2018 AND E1.TID=E2.TID  
AND (E1.LNO<>E2.LNO OR E1.CNO<>E2.CNO ) AND E2.SYEAR=2018;
```

```
SELECT DISTINCT E1.Tid,E1.LNo  
FROM Election AS E1,Election AS E2  
WHERE E1.SYear=2018 AND E1.Tid=E2.Tid  
AND (E1.Lno<>E2.Lno or E1.Cno<>E2.Cno ) AND E2.SYear=2018;
```

100 %

结果 消息

	Tid	LNo
1	001	C01
2	001	c03
3	003	c04
4	003	c05
5	004	c01
6	004	c02
7	005	c01
8	005	c06



练习2

► 查询2018年度讲授过两门或两门以上课程的教师编号和所教授的课程号。

```
SELECT DISTINCT TID,LNO
FROM ELECTION
WHERE SYEAR=2018 AND TID IN
(SELECT TID
FROM ELECTION
WHERE SYEAR=2018
GROUP BY TID HAVING COUNT(*)>=2);
```

```
SELECT DISTINCT TID,LNO
FROM ELECTION
WHERE SYEAR=2018 AND TID IN
(SELECT TID
FROM ELECTION
WHERE SYEAR=2018
GROUP BY TID HAVING COUNT(*)>=2)
```

100 %
结果 消息

	TID	LNO
1	001	C01
2	001	c03
3	003	c04
4	003	c05
5	004	c01
6	004	c02
7	005	c01
8	005	c06



练习3

- ▶ 统计“计算机系”所有教师的教师编号，教师姓名，2018年度教授的总课程数和总学分数，按总学分数从低到高排列。

```
SELECT TEACHER.TID,TNAME,COUNT(ELECTION.LNO),SUM(CREDIT)
FROM TEACHER,ELECTION,LESSON
WHERE TEACHER.TID=ELECTION.TID AND ELECTION.LNO=LESSON.LNO
AND SYEAR=2018 AND DEPARTMENT= '计算机系'
GROUP BY TEACHER.TID,TNAME
ORDER BY 4;
```

```
SELECT Teacher.Tid, Tname, COUNT(election.LNo), SUM(credit)
FROM Teacher, Election, Lesson
WHERE Teacher.Tid=Election.Tid AND Election.LNo=Lesson.LNo
AND syear=2018 AND Department='计算机系'
GROUP BY Teacher.Tid, Tname
ORDER BY 4;
```

	Tid	Tname	(无列名)	(无列名)
1	002	张军	1	3
2	005	丁明	2	4
3	003	王建国	3	7



练习4

- 查询选修了“数据库”但没有选修“软件工程”的班级号，所属专业和该班学生人数。

```
SELECT CLASS.CNO,SPECIALITY,COUNT(SNO)
FROM CLASS,STUDENT
WHERE CLASS.CNO=STUDENT.CNO AND
CLASS.CNO IN (SELECT ELECTION.CNO
FROM ELECTION,LESSON
WHERE LNAME= '数据库' AND ELECTION.LNO=LESSON.LNO)
AND CLASS.CNO NOT IN(SELECT ELECTION.CNO
FROM ELECTION,LESSON
WHERE LNAME=' 软件工程' AND ELECTION.LNO=LESSON.LNO)
GROUP BY CLASS.CNO,SPECIALITY;
```

```
SELECT CLASS.CNO,SPECIALITY,COUNT(SNO)
FROM CLASS,STUDENT
WHERE CLASS.CNO=STUDENT.CNO AND
      CLASS.CNO IN (SELECT ELECTION.CNO
FROM ELECTION,LESSON
WHERE LNAME=' 数据库' AND ELECTION.LNO=LESSON.LNO)
AND CLASS.CNO NOT IN (SELECT ELECTION.CNO |
FROM ELECTION,LESSON
WHERE LNAME=' 软件工程' AND ELECTION.LNO=LESSON.LNO)
GROUP BY CLASS.CNO,SPECIALITY;
```

结果		消息
CNO	SPECIALITY (无列名)	
1	4 计算机	2



练习4

- 查询选修了“数据库”但没有选修“软件工程”的班级号，所属专业和该班学生人数。

```
SELECT CLASS.CNO,SPECIALITY,COUNT(DISTINCT SNO)
```

```
FROM CLASS,STUDENT
```

```
WHERE CLASS.CNO=STUDENT.CNO
```

```
AND CLASS.CNO IN (SELECT CNO
```

```
FROM ELECTION,LESSON
```

```
WHERE LNAME= '数据库'
```

```
AND ELECTION.LNO=LESSON.LNO
```

```
EXCEPT
```

```
SELECT CNO
```

```
FROM ELECTION,LESSON
```

```
WHERE LNAME='软件工程' AND ELECTION.LNO=LESSON.LNO)
```

```
GROUP BY CLASS.CNO,SPECIALITY ;
```

```
SELECT CLASS.CNO, SPECIALITY, COUNT(DISTINCT SNO)
FROM CLASS, STUDENT
WHERE CLASS.CNO=STUDENT.CNO
AND CLASS.CNO IN (SELECT CNO
FROM ELECTION, LESSON
WHERE LNAME='数据库'
AND ELECTION.LNO=LESSON.LNO
EXCEPT
SELECT CNO
FROM ELECTION, LESSON
WHERE LNAME='软件工程' AND ELECTION.LNO=LESSON.LNO)
```

100 %

结果 消息

CNO	SPECIALITY	(无列名)
1	4	计算机
2		



练习5

- ▶ 创建一个视图V1，给出所有“计算机系”学生的学号，姓名，性别，所在班级号和选修“数据库”课程的分。

CREATE VIEW V1

AS SELECT GRADE.SNO,SNAME,GENDER,CNO, SCORE
FROM STUDENT,GRADE,LESSON
WHERE STUDENT.SNO=GRADE.SNO AND LNAME='数据库'
AND GRADE.LNO=LESSON.LNO AND STUDENT.CNO IN
(SELECT CNO FROM CLASS WHERE DEPARTMENT='计算机系')



练习5

- 创建一个视图V1，给出所有“计算机系”学生的学号，姓名，性别，所在班级号和选修“数据库”课程的分。

```
CREATE VIEW V1
```

```
AS SELECT GRADE.SNO,SNAME,GENDER,CLASS.CNO, SCORE
```

```
FROM STUDENT,GRADE,CLASS,LESSON
```

```
WHERE STUDENT.SNO=GRADE.SNO AND
```

```
STUDENT.CNO=CLASS.CNO AND
```

```
DEPARTMENT='计算机系' AND
```

```
GRADE.LNO=LESSON.LNO AND
```

```
LNAME='数据库'
```



练习6

- 查询“计算机系”学生中“数据库”课程分数最高的学生学号，姓名和所得分数。

```
SELECT SNO,SNAME,SCORE  
FROM V1  
WHERE SCORE=(SELECT MAX(Score) FROM V1);
```

```
CREATE VIEW V1  
AS SELECT GRADE.SNO,SNAME,GENDER,CNO, SCORE  
FROM STUDENT,GRADE,LESSON  
WHERE STUDENT.SNO=GRADE.SNO AND LNAME='数据库'  
AND GRADE.LNO=LESSON.LNO AND STUDENT.CNO IN  
(SELECT CNO FROM CLASS WHERE DEPARTMENT='计算机系')
```




小结

-  **SELECT语句实现的查询功能是SQL语言的核心和重点。**
-  **SQL语言可满足用户对数据库的不同查询需求。**
-  **SQL语言是高度非过程化的语言。**